

Chacha Chen

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RESEARCH ☐ **Applied Machine Learning** ☐ **Human Computer Interaction**
KEYWORDS ☐ Reinforcement Learning ☐ Natural Language Processing
 ☐ Healthcare ☐ Crowdsourcing

EDUCATION **Pennsylvania State University** Aug. 2019 - Dec. 2020 (expected)
 College of Information Sciences and Technologies
 Master in Informatics

- **Research Topics:**

- **Human Computer Interaction:** combine AI with crowdsourcing to create systems that are usable, robust, and intelligent.
- **Applied Machine Learning:** explore topics related to applied machine learning in various fields including natural language processing, spatial-temporal data mining, healthcare and etc.

- **Advisor:** Ting-Hao (Kenneth) Huang

Shanghai Jiao Tong University Sep.2015 – Jun.2019
Department of Computer Science
Bachelor of Science in Computer Science
• **Advisor:** Weinan Zhang, Zhenhui (Jessie) Li

Technical University of Munich Apr.2017 – Sep.2017
Exchange Student in Department of Informatics

PUBLICATIONS **Preprints**

1. Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter
Quantitative Analysis of Bibliography Management Applications
To be submitted to CHI Late Breaking Work 2021
2. Jiaqi Wang, Hua Shen, **Chacha Chen**, Frank E. Ritter
Toward an Interactive Online Course Framework
Submitted to CHI Case Studies 2021
3. **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao
A Bayesian Machine Learning Model for Healthcare
Submitted to WWW 2021, under review.
4. Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Zhenhui Li
A Residual Model on Various Tasks
Submitted to AAAI 2021, under review.

Conference Papers

1. Guanjie Zheng, Chang Liu, Hua Wei, Chacha Chen, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**), Chania, Greece, April 2021.
2. Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**), Ghent, Belgium, Sep 2020. (**Best Applied Data Science Paper Award**)
3. **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**), New York, United States, February 2020. (Acceptance rate: $1591/7737=20.6\%$)
4. Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**), Beijing, China, November 2019. (Acceptance rate: $200/1030=19.4\%$)
5. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**), Anchorage, AK, USA, Aug 2019. (Acceptance rate: $170/1200=14.2\%$)

Workshop Papers

1. **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In *Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning* (**NeurIPS 2020**).
2. **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In *Proceedings of the Workshop on Noisy User-generated Text* (**W-NUT, EMNLP 2020**).
3. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, Deep Reinforcement Learning for Traffic Signal Control along Arterials. In *Proceedings of the Workshop on Deep Reinforcement Learning for Knowledge Discovery* (**DRL4KDD, KDD 2019**).

COMMUNITY ACTIVITIES & SERVICE

Conference Program Committee Member/Reviewer

- AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21)
- W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020
- NeurIPS workshop on Interpretable Inductive Biases and Physically Structured Learning

Journal Invited Reviewer

- ACM Transaction on Intelligent System and Technology (TIST)

RESEARCH EXPERIENCE

IQVIA, Analytics Center of Excellence, Boston

Machine Learning Research Intern

May. 2020–Aug. 2020

- Gaussian Process for health risk prediction: Designed Bayesian machine learning models for informative health risk prediction. Different from previous methods, our proposed model is able to give uncertainty-based rich prediction. Moreover, we leveraged multi-sourced data with different modalities.

- Supervised by Cao (Danica) Xiao (IQVIA), Prof. Fenglong Ma (PSU) and Prof. Jimeng Sun (UIUC).

CityBrain Lab, Hangzhou

Machine Learning Research Intern

Jul. 2018–Aug. 2019

- Reinforcement learning for traffic signal control: Designed and implemented a reinforcement learning (RL) approach with justification on state and reward design for multi-intersection traffic signal control
- Supervised by Prof Zhenhui (Jessie) Li (PSU).

SJTU PAMI Lab, Shanghai

Research Assistant

Sep. 2017–Mar. 2018

- Implementation of SVM with indefinite and semi-definite kernels: Coauthored a software package which accommodates Support Vector Machine (SVM) classification algorithms with indefinite kernels and semi-definite kernels. Moreover, we incorporated Sequential Minimal Optimization (SMO) in the traditional SVM algorithm and Provided various kernel functions, including TAHN, TL1, gaussian, polynomial and linear kernels
- Supervised by Prof Xiaolin Huang (SJTU).

TEACHING EXPERIENCE	Pennsylvania State University	
	Teaching Assistant IST 261 (Java Course): Application Development Design Studio I Prof. Kenneth Huang	Fall 2020
AWARDS	AAAI-20 VISA Research Scholarship	Feb. 2020
	AAAI-20 Student Travel Grant	Feb. 2020
	Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i>	Aug. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016
RELEVANT SKILLS	Technical: Python, C++, Database, Java, Matlab, R	